

Linear Regression & Gradient Descent | Lecture 2

Predicting house prices



Learning algorithm

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h (hypothesis)

How to represent h?

$$h(x) = \theta_0 + \theta_1 x_1 + \theta_2 x_2$$

x_1 = size of house

x_2 = # bedrooms

$$h(x) = \sum_{j=0}^2 \theta_j x_j$$

where $x_0 = 1$

$$\theta = \begin{bmatrix} \theta_0 \\ \theta_1 \\ \theta_2 \end{bmatrix}$$

↓
Parameters

$$x = \begin{bmatrix} x_0 \\ x_1 \\ x_2 \end{bmatrix}$$

$m \rightarrow$ no. of training

[no. of rows in the training example]

X = "inputs / features"

y = "output / target variable"

(x, y) = training example.

$x^{(i)}, y^{(i)}$ = i^{th} training example

$n \rightarrow$ features

Choose θ such that $h(x) \approx y$
for training examples.

$h_{\theta}(x) = h(x) \rightarrow$ both mean the same.

For linear regression $\Rightarrow$

minimize $\rightarrow (h_{\theta}(x) - y)^2$

minimize $\frac{1}{2} \sum_{i=1}^m (h_{\theta}(x^{(i)}) - y^{(i)})^2$

minimize $J(\theta) = \frac{1}{2} \sum_{i=1}^m (h_{\theta}(x^{(i)}) - y^{(i)})^2$

Gradient descent

start with some value of θ ($\theta = \theta_0$) &
keep changing θ until it minimizes $J(\theta)$.

$$\theta_j := \theta_j - \alpha \frac{\partial}{\partial \theta_j} J(\theta)$$

$\alpha \rightarrow$ learning rate

$$\frac{\partial}{\partial \theta_j} J(\theta) = (h_{\theta}(x) - y) x_j$$

$$\sum_{i=1}^m \theta_j = \theta_j - \alpha (h_{\theta}(x) - y) x_j$$

$$\theta_j = \theta_j - \alpha \sum_{i=1}^m (h_{\theta}(x^{(i)}) - y^{(i)}) x_j^{(i)}$$

Repeat until convergence.

The disadvantage of a batch gradient descent

~~The alg~~

Stochastic Gradient Descent.

Repeat

for $i=1$ to m

$\downarrow$

for every j

$$\theta_j = \theta_j - \alpha (h_{\theta}(x^{(i)}) - y^{(i)}) x_j^{(i)}$$

change θ to $\theta + \Delta \theta$

Normal Equation works only for linear regression

$$\nabla_{\theta} J(\theta)$$

$$\theta \in \mathbb{R}^{n+1}$$

$$= \begin{bmatrix} \frac{\partial J}{\partial \theta_0} \\ \frac{\partial J}{\partial \theta_1} \\ \frac{\partial J}{\partial \theta_2} \end{bmatrix}$$

$$A \in \mathbb{R}^{2 \times 2}$$

$$A = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix}$$

$$f(A) = A_{11} + A_{12}^2$$

$$f\left(\begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix}\right) = 5 + 6^2$$

$$\nabla_A f(A) = \begin{bmatrix} \frac{\partial f}{\partial A_{11}} & \frac{\partial f}{\partial A_{12}} \\ \frac{\partial f}{\partial A_{21}} & \frac{\partial f}{\partial A_{22}} \end{bmatrix} = \begin{bmatrix} 1 & 2A_{12} \\ 0 & 0 \end{bmatrix}$$

If A is a square matrix ($A^{n \times n}$)

$$\text{tr} A = \text{sum of diagonals}$$

$$\text{tr} A = \text{tr} A^T$$

Proposition of trace matrix. $\rightarrow B$ is a fixed matrix

$$f(A) = \text{tr } AB$$

$$\nabla_A f(A) = B^T$$

$$\text{tr } AB = \text{tr } BA$$

$$\text{tr } ABC = \text{tr } CAB$$

$$\nabla_A \text{tr } AA^T C = CA + C^T A$$

$$J(\theta) = \frac{1}{2} \sum_{i=1}^m (h(x^{(i)}) - y^{(i)})^2$$

$$X\theta = \begin{bmatrix} (x^{(1)})^T \theta \\ (x^{(2)})^T \theta \\ \vdots \\ (x^{(m)})^T \theta \end{bmatrix} = \begin{bmatrix} \theta_0 \\ \theta_1 \\ \theta_2 \end{bmatrix}$$

$$= \begin{bmatrix} x^{(1)T} \theta \\ x^{(2)T} \theta \\ \vdots \\ x^{(m)T} \theta \end{bmatrix} = \begin{bmatrix} h_\theta(x^{(1)}) \\ \vdots \\ h_\theta(x^{(m)}) \end{bmatrix}$$

$$\vec{y} = \begin{bmatrix} y^{(1)} \\ y^{(2)} \\ \vdots \\ y^{(m)} \end{bmatrix}$$

$$J(\theta) = \frac{1}{2} (X\theta - y)^T (X\theta - y)$$

$$\nabla_{\theta} J(\theta) = \nabla_{\theta} \frac{1}{2} (X\theta - y)^T (X\theta - y) = 0 \quad (\text{objective})$$

$$= \frac{1}{2} \nabla_{\theta} (\theta^T X^T - y^T) (X\theta - y)$$

$$= \frac{1}{2} \nabla_{\theta} [\theta^T X^T X\theta - \theta^T X^T y - y^T X\theta + y^T y] = 0$$

$$= \frac{1}{2} [X^T X\theta + X^T X\theta - X^T y - X^T y] = 0$$

$$X^T X\theta = X^T y$$

$$\theta = (X^T X)^{-1} X^T y$$